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Retaining arrangement for an optical element

Abstract

The disclosure relates to a retaining arrangement with a carrier platform (2), for example in laser systems, to which at least one optical element (3) is fixed. The disclosure specifies a retaining arrangement for optical elements which ensures improved beam position stability with as little effort as possible. For this purpose, the carrier platform (2) is connected to a base (8) at bearing points (5) via a respective elastically compliant and/or damping connecting structure (6). The connecting structure (6) is designed to elastically absorb thermal expansions of the carrier platform (2). The at least one optical element (3) is located at a neutral point (13) and/or on a neutral axis on the carrier platform (2), wherein this neutral point (13) or the neutral axis is positionally stable relative to the base (8) during thermal deformation of the carrier platform (2). The carrier platform (2) is connected to a heat sink or source (16) via an intermediate heat pump, for example in the form of a Peltier element (17), with a flexible, for example ribbon-shaped heat transfer element (15) without mechanical retroaction.

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Background/Summary

FIELD AND BACKGROUND OF THE INVENTION

- (1) The present disclosure relates to a retaining arrangement with a carrier platform to which at least one optical element is fixed.
- (2) In optical systems, mounting concepts are required for the optical elements used (e.g. lenses, filters, prisms, etc.). In laser systems, there is often a requirement to fix optical subsystems to a base (screw-on surface) of the laser system, whereby an optical axis of the respective subsystem should be as positionally stable as possible relative to the base.
- (3) For example, the direction of the output beam from a laser system is subject to certain variations that can cause significant problems, such as when the beam must be coupled into a light-conducting fiber or out of a light-conducting fiber and/or when the beam must accurately strike a target at a greater distance.
- (4) Occurring beam direction fluctuations can have various causes. If the laser beam propagates through an optical subsystem connected to the laser system, this generally has an influence on the beam guidance, whereby this influence strongly depends on how positionally stable the optical elements of the subsystem are relative to the base. For example, if laser radiation emerges from one end of an optical fiber in a coupling unit attached to the laser system and is collimated by a lens, the position of the laser beam downstream of the lens will exhibit undesirable transverse motion corresponding to occurring angular variations of the lens. It is desirable to have a design in which the fluctuations in the angular orientation of the laser beam are only a small fraction of the beam divergence.
- (5) It is known to minimize beam position fluctuations by minimizing mechanical vibrations of the relevant optical elements through a stable design and mechanical decoupling. As a further measure to avoid beam position fluctuations, it is known to minimize thermal drifts by shielding heated

components such as laser diodes or electronic circuits from the optical elements influencing the beam position for example.

(6) In practice, the known approaches often prove to be insufficient or too complex and thus too expensive for certain applications.

SUMMARY OF THE INVENTION

(7) The present disclosure relates to a retaining arrangement of the type indicated at the outset in that the carrier platform is connected to a base at bearing points in each case via an elastically compliant and/or damping connecting structure, the connecting structure being designed to elastically absorb thermal expansions of the carrier platform, the at least one optical element being located at a neutral point on the carrier platform, this neutral point being characterized as such by the fact that it is positionally stable relative to the base in the event of thermal deformation of the carrier platform.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1*a,b*: shows a schematic top view on a retaining arrangement from above;

(2) FIG. 2: shows a schematic side view on the retaining arrangement of FIG. 1.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(3) The present disclosure relates to a retaining arrangement of the type indicated at the outset in that the carrier platform is connected to a base at bearing points in each case via an elastically compliant and/or damping connecting structure, the connecting structure being designed to elastically absorb thermal expansions of the carrier platform, the at least one optical element being located at a neutral point on the carrier platform, this neutral point being characterized as such by the fact that it is positionally stable relative to the base in the event of thermal deformation of the carrier platform. The carrier platform is connected to a heat sink or source via an intermediate heat pump, for example in the form of a Peltier element, with a flexible, for example ribbon-shaped heat transfer element.

(4) The base represents the reference structure for the retainer, as it were. Via the base, for example, the optical subsystem arranged on the carrier platform can be connected to a laser system.

Likewise, the base can, for example, be part of a housing or a base frame of a laser system or another optical system.

(5) The disclosure is based on providing mechanical decoupling via the elastically compliant and/or damping connection structure, so that influences of mechanical fluctuations on the optical elements fixed on the carrier platform are minimized. At the same time, thermal influences are minimized by positioning the at least one optical element relevant for the beam position at the neutral point and/or on the neutral axis on the carrier platform. The neutral point is defined by the fact that it remains positionally stable relative to the base in the event of thermal deformation of the carrier platform, e.g. due to thermal expansion in the event of temperature fluctuations.

(6) In one possible embodiment, the carrier platform forms an optical bench, wherein an optical axis defined by the at least one optical element fixed on the carrier platform (2) passes through the neutral point. In this way, the desired beam stability can be achieved.

(7) The bearing points are conveniently positioned in a statically suitable arrangement for the retaining task. In one embodiment, for example, the bearing points can be arranged spaced apart from one another on the outer circumference of the carrier platform. In principle, however, any arrangement of the bearing points on the carrier platform is conceivable. The number of bearing points can also vary. The support via the bearing points serves to ensure that no undesirable tilting movements are possible. For example, three (or more) bearing points can be provided, which do not all lie on a common axis.

- (8) In the case of thermal deformations, an elastic displacement occurs in the compliant connection structures. With the most diverse conceivable geometries of the carrier platform with the most diverse arrangements of the bearing points on the carrier platform, a neutral point or a neutral axis can be found which always remains positionally stable under thermal deformations. According to the disclosure, the optical element critical for the beam position is arranged there.
- (9) In other words, the arrangement of the bearing points compensates for quasi-static movements due to changes in the isotropic ambient temperature field by absorbing these movements through the elastic compliance of the connecting structure. This compliance allows a “sacrificial movement” to be performed, so to speak, which ensures that the optical element located at the neutral point remains positionally stable and that the beam position accordingly does not change (i.e. only within tolerable limits) despite temperature fluctuations occurring.
- (10) The approach to beam position stabilization according to the disclosure can be further supported by selecting as material for the carrier platform a material with thermally/mechanically invariant behavior as far as possible, i.e. a material or material composition with thermal expansion as low as possible.
- (11) The carrier platform is connected to a heat sink or source via a flexible, for example ribbon-shaped heat transfer element. In some applications, there is a requirement to temper the optical elements located on the carrier platform, i.e., to bring them to a certain operating temperature and keep them there. According to the disclosure, the heat conduction path required for this purpose is formed by the flexible heat transfer element, for example in the form of a flexible ribbon of metal mesh. The flexibility of the heat transfer element provides that the thermal connection to the heat sink or heat source is mechanically without retroaction, meaning that the mechanical decoupling provided by the connection structure is not impaired by the thermal connection. In addition, the flexible heat transfer element offers the possibility that the heat sink or source connected above it can be positioned freely, for example also at a greater distance from the carrier platform. When designing the heat transfer element, care should be taken to ensure that the connection to the carrier platform covers an area as large as possible. Temperature control can be achieved by specifying the heat flow via the heat transfer element.
- (12) The carrier platform is connected to the flexible heat transfer element via an intermediate heat pump, for example in the form of a Peltier element. Temperature control can be realized by suitable control of the Peltier element, ideally in combination with temperature measurement on the carrier platform. The Peltier element thereby controls the heat flow from the heat source to the carrier platform or from the carrier platform to the heat sink. Thus, the carrier platform and the optical elements arranged thereon can be kept stable at a predetermined temperature. In one possible embodiment, the connecting structure includes thermal insulation provided to reduce or substantially eliminate heat flow between the carrier platform and the base. Through the thermal insulation thermal influences on the carrier platform and the optical elements located thereon are minimized.
- (13) In one practical embodiment, the connecting structure may comprise a connecting member connected to the base and connected to the carrier platform via an elastic solid joint. An elastic solid joint, for example in the form of a plate joint, offers the possibility that the displacement motion of the connecting member relative to the carrier platform is well defined. For example, a plate joint can be designed to be compliant along only one axis. By combining such joints at the various bearing points with specific alignment of the respective displacement axes, the position of the neutral point or neutral axis on the carrier platform can be specifically specified to meet the requirements.
- (14) In a simple realization, the connecting member can be a bolt screwed to the base and guided in a sleeve (e.g. threaded sleeve) connected to the carrier platform via the elastic solid joint. Due to the reduced material cross-sections in the area of the joint, the solid joint also offers the possibility that it intrinsically reduces the heat flow from the base to the carrier platform via the connecting

structure (thermal insulation).

(15) The solid joint can also be designed to provide sufficient damping for the desired mechanical decoupling.

(16) In one possible embodiment, the at least one optical element is an element from the list: Lens, optical grating, prism, beam splitter, filter, end of an optical fiber, laser diode, or a combination of these elements, to name just a few exemplary possibilities.

(17) In an application, an end of a light conducting fiber and a lens are fixed to the carrier platform, with the lens located at the neutral point. In this case, the retaining arrangement holds the components of a fiber coupler for coupling light out of the fiber or coupling light into the fiber. The beam position stabilization achieved according to the disclosure can ensure efficient and stable outcoupling or incoupling.

(18) In the following figure description, the same reference signs and the same terms are used for the same elements.

(19) In the figures, the retaining arrangement as a whole is designated by the reference numeral **1**. It has a carrier platform **2** in the form of a polygonally shaped block in plan view (FIG. **1**). A focusing lens **3** and one end **4** of a light-conducting fiber are fixed to the carrier platform **2** as optical elements. The focusing lens **3** focuses a laser beam **7** coming from the left in the drawings onto the fiber end **4** to couple the radiation into the light-conducting fiber. The carrier platform **2** is connected to a base **8** (FIG. **2**) at three bearing points **5** distributed over the outer circumference of the carrier platform **2** in a triangular configuration, each via an elastically compliant and damping connection structure **6**.

(20) The connecting structures **6** each comprise a connecting member connected to the base **8** in the form of a screw bolt **9** (FIG. **2**), which is screwed into the base **8**. The screw bolt **9** is guided in each case in a sleeve **10**, which is connected to the carrier platform **2** via a solid joint **11** designed as a plate joint. The plate joints each define a deflection axis **12**. By combination of the joints **11** at the three bearing points **5** with the illustrated alignment of the deflection axes **12** the position of an excellent point **13** on the carrier platform **2** (FIG. **1**) is specified. When thermal expansion occurs due to temperature fluctuations, elastic displacement occurs along the axes **12**. The thermal expansion is thereby absorbed by the solid joints **11**. In the geometry shown, the excellent point **13** results as a neutral point which always remains positionally stable in relation to the base **8** during thermal deformations. Accordingly, the lens **3** is arranged in the point **13** so that the position of the lens **3** and thus the focus of the laser beam **7** on the fiber end **4** remains unchanged stable during temperature fluctuations.

(21) With the geometry of the bearing points **5** in combination with the three solid joints **11**, the neutral point **13** results, as explained. In the case of a deviating geometry, in which, for example, the deflection axes **12** do not intersect at a point but run parallel to each other, a neutral axis (not shown) can alternatively be realized.

(22) The connection structure **6** comprises thermal insulation in the form of spacer sleeves **14** made of thermally insulating material, which are provided to minimize heat flow between the carrier platform **2** and the base **8**.

(23) As can be seen in FIG. **2**, the carrier platform **2** is connected to the base **8** only via the screw bolts **9** of the connecting structure **6** with the spacer sleeves **14** arranged between them. Otherwise, there is no material contact between the carrier platform **2** and the base **8**.

(24) The carrier platform **2** is connected to a heat sink or source **16** via a flexible, ribbon-shaped heat transfer element **15** (FIG. **2**). A heat conduction path is formed by the flexible heat transfer element **15** in order to be able to selectively temper the carrier platform **2** with the optical element **3** arranged thereon. Due to the flexibility of the heat transfer element **15**, the thermal connection to the heat sink or source **16** is mechanically without retroaction. The mechanical decoupling provided by the compliant connection structure **6** is thus not impaired by the thermal connection. The carrier platform **2** is connected to the flexible heat transfer element **15** via an intermediate

Peltier element **17**. By controlling the Peltier element, temperature control or regulation can be implemented. The Peltier element **17** controls the heat flow from the heat source **16** to the carrier platform **2** or from the carrier platform **2** to the heat sink **16**.

(25) The disclosure specifies a retaining arrangement for optical elements that ensures improved beam position stability with as little effort as possible.

Claims

1. A retaining arrangement having a carrier platform to which at least one optical element is fixed, wherein the carrier platform is connected to a base at bearing points via a respective elastically compliant and/or damping connecting structure, wherein the connecting structure is designed to elastically absorb thermal expansions of the carrier platform, wherein the at least one optical element is located at a neutral point and/or on a neutral axis on the carrier platform, wherein this neutral point and/or neutral axis is positionally stable relative to the base during thermal deformation of the carrier platform, wherein the carrier platform is connected to a heat sink or source via an intermediate heat pump with a flexible heat transfer element that provides a thermal connection without mechanical retroaction, thereby maintaining the mechanical decoupling provided by the connecting structure.
 2. The retaining arrangement according to claim 1, wherein the bearing points are arranged at a distance from each other on the carrier platform.
 3. The retaining arrangement according to claim 1, wherein the connection structure comprises a thermal insulation provided to reduce a heat flow between the carrier platform and the base.
 4. The retaining arrangement according to claim 1, wherein the connecting structure comprises a connecting member (**9**) connected to the base, which is connected to the carrier platform via an elastic solid joint.
 5. The retaining arrangement according to claim 4, wherein the connecting member is a bolt screwed to the base, which is guided in a sleeve, which is connected to the carrier platform via the elastic solid joint.
 6. The retaining arrangement according to claim 1, wherein the carrier platform forms an optical bench, wherein an optical axis defined by the at least one optical element fixed on the carrier platform passes through the neutral point.
 7. The retaining arrangement according to claim 1, wherein the at least one optical element is an element selected from the list: Lens, optical grating, prism, beam splitter, filter, end of a light conducting fiber, laser diode.
 8. The retaining arrangement according to claim 1, wherein an end of a light conducting fiber and a lens are fixed to the carrier platform, wherein the lens is located at the neutral point.
 9. A laser system comprising a retaining arrangement according to claim 1, wherein at least one optical element of the laser system is fixed to the retaining arrangement.
 10. The retaining arrangement according to claim 1, wherein the intermediate heat pump is a Peltier element.
 11. The retaining arrangement according to claim 1, wherein the flexible heat transfer element is a ribbon-shaped element.
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